

## Chapter 153 — Epidemiology, Aetiology and Prevention of Prostate Cancer — Numbers, Gold Standards and Traps

### IF YOU READ NOTHING ELSE

1. **Established risk factors:** age, Black ancestry, family history — the only unequivocal three.
2. **First-degree relative** → 2–3×; father + brother → 9×.
3. **Heritability** ≤58% (twins); BRCA2, BRCA1, ATM, PALB2, CHEK2 (homologous recombination repair) → aggressive disease.
4. **ERSPC (European Randomized Study of Screening for Prostate Cancer):** screening ↓ prostate cancer death ~20% (intention to screen) → 27–31% adjusted.
5. **PSA (prostate-specific antigen) screening** = strongest predictor of diagnosis → healthy-user bias.
6. **Obesity:** ↓ low-grade, ↑ high-grade/fatal disease.
7. **Smoking:** incidence ↔ / ↓, fatal ↑; quit ≥10 yr → excess gone.
8. **Diabetes** → ↓ risk (↓ testosterone, IGF-1 [insulin-like growth factor 1], PSA; metformin).
9. **PCPT (Prostate Cancer Prevention Trial)/REDUCE (REduction by DUtasteride of prostate Cancer Events):** 5ARI (5α-reductase inhibitor) ↓ cancer ~25%, ↑ high-grade; mortality null.
10. **SELECT (Selenium and Vitamin E Cancer Prevention Trial):** selenium/vitamin E — no benefit; vitamin E alone trend ↑.

### EPIDEMIOLOGY AND AETIOLOGY

- **Worldwide:** 2nd commonest non-skin cancer; 5th cause of male cancer death.
- **2020:** ~1.4 million cases, 375,000 deaths; global ASR (age-standardised rate) 31/100,000; lifetime risk 3.9%.
- **First for incidence** in 112/185 countries; 2nd cause of cancer death in 48.
- **HDI (Human Development Index):** high 37.5 vs low 11.3 per 100,000.
- **Incidence range:** 6.3 South Central Asia → 83.4 Northern Europe.
- **Mortality range:** 3.1 South Central Asia → 27.9 Caribbean (highest).
- **Northern America:** incidence 73, mortality only 8.3.
- **Europe:** commonest cancer; 3rd cause of cancer death; Poland only rising mortality.
- **USA:** lifetime risk 1 in 8 (12.6%); 29% of male cancers.
- **USA:** 5-yr survival 97% yet 2nd cause of male cancer death.
- **USA age risk:** 0.2% <50; 1.8% 50s; 5.2% 60s; 9.2% ≥70.
- **Black US males:** 37% of cancer diagnoses, 17% of cancer deaths.
- **Canada:** lifetime 1 in 9; incidence peaks 1993 (PSA), 2011.
- **Caribbean highest national incidence:** Martinique 182, Guadeloupe 177.
- **Hispanic paradox:** worse stage/grade/PSA, lower mortality.
- **Asia:** lowest incidence and mortality; highest Japan, Israel, Armenia.
- **Asian migrants:** native Asian risk one-third lower than Asian American.
- **Oceania:** Australia/NZ 75.8; highest New Caledonia.
- **Preferred end point:** aggressive/fatal disease, not incidence.

### PROSTATE-SPECIFIC ANTIGEN SCREENING

- **PSA raised by:** BPH (benign prostatic hyperplasia), inflammation, cancer — anything increasing leakage.
- **ERSPC excess incidence:** 85% (9 yr) → 60% → 49% → 36% (16 yr).
- **ERSPC mortality:** ↓ 16% (9 yr) → 22% (11 yr) → 20% (16 yr).
- **Adjusted:** non-attendance → 27%; + contamination → 31%.
- **PLCO (Prostate, Lung, Colorectal and Ovarian trial):** contaminated control arm vs ERSPC clean.
- **Autopsy latent prevalence:** 5% <30 → 59% >79.
- **Age 70–79 latent:** White 36% vs Black 51%.

### RISK FACTORS

- **Age:** 60% diagnosed ≥65; mean 66; 55% of deaths ≥65.
- **Young adults (15–39):** incidence rising ~2%/yr.
- **Black males:** highest incidence (172.6/100,000 US); mortality 2×.
- **Equal access** → equal cancer-specific mortality (non-metastatic).
- **Family history:** 20% of patients.
- **Relative diagnosed <60** → higher risk than ≥60.
- **Father/brother died** of prostate cancer → 2× death risk.
- **8q24:** commoner in African ancestry.
- **17q21:** 4–7% African vs <1% non-African.
- **HOXB13:** prostate-specific; hereditary disease.
- **ATM:** metastatic disease; **PALB2:** aggressive, lethal.
- **MMR (mismatch repair; Lynch syndrome):** MLH1, MSH2, MSH6, PMS2; 2–5%; ADT (androgen deprivation therapy)-sensitive.
- **PARP (poly[ADP-ribose] polymerase) inhibitor:** BRCA1/2, ATM; **checkpoint inhibitor:** MMR.
- **GWAS (genome-wide association studies):** >200 SNPs (single-nucleotide polymorphisms); KLK3 (PSA gene) SNPs.
- **East Asian:** ERG-positive rare; FOXA1 mutated in Chinese.

### POSSIBLE RISK FACTORS

- **Height:** ~5% ↑ per 10 cm; high-grade/fatal only.
- **Tall + Black:** predicts progression; not in White males.
- **Weight gain after prostatectomy** → ↑ recurrence.
- **Metabolic syndrome:** ↑ oestradiol, ↓ SHBG (sex hormone-binding globulin), ↓ free testosterone.
- **Metabolic syndrome:** ↑ Gleason, positive margins, biochemical failure.
- **Hypertension:** systolic ≥150 vs <130 mmHg → ↑ risk, ↑ death.
- **ADT:** randomised-trial meta-analysis — no ↑ cardiovascular death.
- **Cardiovascular disease:** leading cause of death in prostate cancer patients.

### DIET

- **WCRF/AICR (World Cancer Research Fund/American Institute for Cancer Research) 2018:** no dietary factor reaches strong evidence.
- **Downgraded 2007 → 2018:** dairy, calcium, lycopene.
- **Fat:** null in PSA era.
- **Sugar-sweetened drinks** ↑; glycaemic index null.
- **Heterocyclic amines:** meat >130 °C; aggressive disease only.

- **Lycopene:** cooked in oil; ↓ lethal cancer in both eras.
- **Coffee:** ↓ advanced/fatal; decaffeinated same.
- **Alcohol:** ≥45 g/day → ↑ mortality; liquor > wine.
- **Binge drinking** ↑ total risk.
- **Plant-based (HPFS, Health Professionals Follow-up Study):** ↓ fatal — avoiding animal foods.

## TOBACCO, ACTIVITY AND LIFESTYLE

- **Smoking:** ~40% ↑ prostate cancer mortality; Surgeon General acknowledged.
- **Vigorous (not moderate) activity:** ↓ advanced (not total) disease.
- **Sedentary:** ↑ aggressive disease via adiposity.
- **Healthy lifestyle (HPFS/PHS, Physicians' Health Study):** ↓ lethal disease 30–50%.
- **Guideline adherence:** prevents 21% of fatal cancers.
- **Top PRS (polygenic risk score) quartile:** lethal risk 5.3% unhealthy vs 1.6% healthy.

## SUPPLEMENTS, VITAMINS AND MINERALS

- **USPSTF (US Preventive Services Task Force):** only multivitamins — small ↓ cancer incidence.
- **VITAL (Vitamin D and Omega-3 Trial):** D3 2000 IU — prostate null; ↓ metastatic/fatal cancer overall.
- **SELECT:** 35,533; stopped early at 5.46 yr.
- **ATBC (Alpha-Tocopherol, Beta-Carotene study):** α-tocopherol ↓ incidence 32%, fatal 41%.
- **ATBC vs SELECT:** smokers, 10× lower dose, no PSA screening.
- **Calcium supplements:** ↑ fatal disease → use cautiously.
- **Zinc ≥75 mg/day or ≥15 yr** → ↑ lethal, high-grade.
- **High-dose antioxidants:** ↑ advanced, high-grade.
- **Green tea:** significant only >7 cups/day.
- **Soy protein:** randomised trial — no effect on recurrence.
- **Pomegranate:** placebo-controlled trial null.

## ASSORTED RISK FACTORS

- **Transgender women on GAHT (gender-affirming hormone therapy):** median PSA 0.02 vs 1 ng/mL.
- **Transgender women:** incidence lower, disease more aggressive.
- **Vasectomy:** AUA (American Urological Association) — not a risk factor; no counselling needed.
- **Ejaculation:** ≥4 to ≥21/month → ↓ risk.
- **HPV (human papillomavirus):** modest ↑ (~28%) vs 95% of cervical cancer.
- **Shift work:** IARC (International Agency for Research on Cancer) 2007 "probably carcinogenic".
- **Organochlorines (DDT, dichlorodiphenyltrichloroethane), chromium:** modest ↑.
- **PFAS (per- and polyfluoroalkyl substances), cadmium:** no link.
- **Early puberty** → ↑ aggressive disease (Mendelian randomisation).
- **Free (not total) testosterone** (Mendelian randomisation) ↑; <60 only.
- **Vertex baldness** (only) → ↑ risk.
- **Insulin, IGF-1** → ↑ risk.

## PRIMARY PREVENTION

- **No FDA (US Food and Drug Administration)-approved** chemopreventive agent.
- **Bar:** must prevent aggressive, not total, cancer.
- **5ARI:** prostate volume ↓ ~25%, PSA ↓ ~50%.
- **PCPT:** finasteride 5 mg; PSA ≤3.0; 7 yr; 18,882.
- **PCPT:** cancer 18.4% vs 24.4% (↓ 24.8%).
- **PCPT:** Gleason 7–10: 6.4% vs 5.1%.
- **PCPT at 18.4 yr:** non-significant 25% ↓ death.
- **REDUCE:** dutasteride 0.5 mg; PSA 2.5–10 + negative biopsy.
- **REDUCE:** 19.9% vs 24.9% (↓ 22.8%); Gleason 8–10 0.9% vs 0.6%.
- **FDA:** rejected artefact explanations; high-grade warning on label.
- **5ARI observational:** ↓ diagnosis; mortality null.
- **Aspirin/NSAIDs (non-steroidal anti-inflammatory drugs):** small ↓ incidence; mortality null; not routine.
- **Statins:** ↓ advanced/fatal (observational); no randomised trial.
- **Metformin:** null.

## SECONDARY PREVENTION

- **Lifestyle trial (surveillance):** treatment 27% control vs 5% intervention.
- **Activity:** ~31% ↓ cancer-specific mortality.
- **Guidelines:** ≥150 min/wk moderate or ≥75 min/wk vigorous.
- **Whole milk ≥4 servings/wk** → ↑ recurrence, mortality.
- **Western diet (PHS):** HR 2.53 cancer-specific, 1.67 all-cause mortality.
- **Prudent diet:** 36% ↓ all-cause mortality.
- **Quit ≥10 yr** → recurrence as never-smoker.
- **≥2 drinks/day** (liquor) → ↑ cancer-specific mortality.
- **Pomi-T:** ↓ PSA rise (surveillance, recurrence).

## CLASSIC TRAPS

- **Trap:** high incidence = high mortality — **Actually:** Northern America 73/8.3; Caribbean highest mortality.
- **Trap:** healthy habits raise risk — **Actually:** healthy users get PSA tests.
- **Trap:** smoking ↑ incidence — **Actually:** incidence ↔ / ↓; fatal disease ↑.
- **Trap:** diabetes ↑ risk as for other cancers — **Actually:** ↓ prostate cancer.
- **Trap:** obesity ↑ all prostate cancer — **Actually:** ↓ low-grade, ↑ high-grade/fatal.
- **Trap:** 5ARIs ↑ cancer death — **Actually:** ↑ high-grade detection; mortality null.
- **Trap:** PRS predicts aggressive disease — **Actually:** predicts any cancer only.
- **Trap:** vasectomy is a risk factor — **Actually:** detection bias; AUA: no.
- **Trap:** "normal" PSA reassures on GAHT — **Actually:** PSA ~50× lower.
- **Trap:** adult serum testosterone ↑ risk — **Actually:** null; genetic free testosterone ↑.